Regis University CPS
CS361 Programming Assignment 2
due by midnight Sunday of week 2

The topic of this assignment was chosen so that you won't allow mathematical equations and scientific notation to intimidate you. No knowledge is required to complete the assignment beyond knowing how to perform mathematical operations within a formula.

> NOTE: Use the **double** precision data type for all ***floating point*** variables in these programs (instead of the single precision **float** data type).

Program #1

Electro-magnetic force describes the influence that one electrical charge has on another when neither charge is moving.

The equation for this **electro-magnetic force** in Newtons is:

$$F = K\,\frac{q_1 q_2}{r^2}\ \text{, where the constant } K = \frac{1}{4\pi\varepsilon_0}$$

and where:

$\boldsymbol{\pi}$ is the constant Pi, whose value to 10 decimal places is 3.1415926535

$\mathbf{E_0}$ is a constant known as the ***permittivity of free space***
with value = 8.854×10^{-12} coulombs^2 per Newton meters^2

$\mathbf{q_1}$ and $\mathbf{q_2}$ represent the charges of two objects in coulombs

r represents the distance between the centers of the objects in meters

K is ***Coulomb's constant***, in Newton meters^2 Coulombs^{-2}

For the Curious: Electrical charges can either attract or repel each other, as $\mathbf{q_1}$ and $\mathbf{q_2}$ may be either positive or negative. When the have same signs, they repel each other. When they have opposite signs, they attract each other.

Program Requirements:

Write a program to compute the electro-magnetic force between 2 charged objects.
The program shall:

- Define a constant, using ***scientific*** notation, for the ***permittivity of free space.***
- Define another constant, using ***fixed*** notation, for ***Pi*** carried out to 10 decimal places.

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- Use a separate, **user-defined function** to output the program description header to the user before prompting for input. This header should contain **your name**.

  For example:

```
Calculation of Electro-magnetic Force between two Objects
                        By
                   Mr. Joe Science
```

- Read user input

  NOTE: You are not required to validate any of the user input. You may assume that the user will enter a valid answer for each question.

  - Read the charges of two objects, as input from the user.
  - Read the distance between the centers of the two objects, as input from the user.
- Calculate the electro-magnetic force between the objects.
- Display several blank lines after reading all the inputs, before displaying the results.
- Displays all data using the format given in the following sample output, which meet the following specifications:
  - ✓ The charges of both objects, and the distance between the objects, are displayed first, in **fixed** format, carried out to 3 decimal places. Their decimals should all line up.
  - ✓ The permittivity of free space, the calculated Coulomb's constant, and the calculated electro-magnetic force are 1ll displayed in **scientific** format, carried out to 4 decimal places. Their decimals should line up.
  - ✓ All ***unit descriptions*** also line up.

Additionally, the program must:

- Compile without errors
- Include appropriate mnemonic constant and variable declarations
- Prompt the user for data in a meaningful way
- Adhere to the documentation standards for this course (see course Content section 1.8)

(see sample input/output on next page)

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Sample Input and Output:

```
Calculation of Electro-magnetic Force between two Charged Objects
                        By
                  Mr. Sam Electric

Enter charge of object 1 (in coulombs): 1.1
Enter charge of object 2 (in coulombs): 2.2

Enter distance between the centers of the two objects
(in meters): 3.33



RESULTS:

  Charge of first object:              1.100 coulombs
  Charge of second object:             2.200 coulombs
  Distance between objects             3.330 meters

  Permittivity of Free Space:    8.8540e-012 coulombs^2 per Newton meters^2
  Coulomb's Constant:            8.9877e+009 Newton meters^2 Coulombs^-2

  Electromagnetic Force:         1.9615e+009 Newtons
```

Notes:

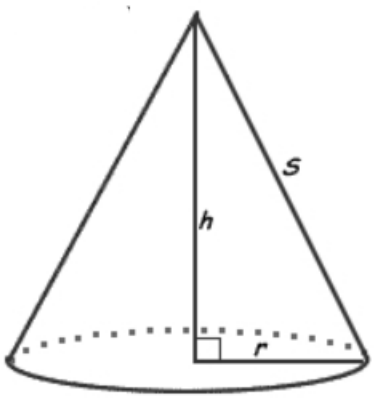
- This output example provides convenient test data that can be used to verify the calculations in your program.
- The representation of the following number uses scientific notation:
    - 8.854×10^{-12} is the same as 0.000000000008854, and is represented in C++ scientific notation as 8.854E-12.
    - The minus sign in 8.854E-12 indicates a negative exponent (i.e. a fraction).
    - By default, the exponent in scientific notation is displayed using 3 digits, like this: 8.854E-012.
    - The number of digits in the scientific notation can be changed via the **setprecision** output manipulator (i.e. the same method as **fixed** notation).
- Users can also input values at the keyboard using scientific notation

(see next page for program 2 requirements)

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Program #2

A **right circular cone** has a circular base, and the line from the center of the base to the tip of the cone forms a right angle with the base.



If you know the **height (h)** of a cone and the **radius (r)** of the base of the cone, then you can calculate the **slant height (s), surface area (a),** and **volume (v)** of the cone using the following formulas.

Slant Height: $s = \sqrt{h^2 + r^2}$

Surface Area: $A = \pi r^2 + \pi r s$

Volume: $v = \frac{\pi r^2 h}{3}$

Note that the slant height is the hypotenuse of the right triangle, whose other two sides are the height and the radius of the base.

Again, don't let the equations intimidate you. These formulas simply perform addition, multiplication, division, squaring, and finding a square root.

Requirements:

Write a program that will:

- Display a program description to the user before asking for any input.
- Defines a constant for PI, using ***fixed*** notation, carried out to 5 decimal places (value for Pi was given in program 1).
- Read the height of a right circular cone, as input from the user.
- Read the radius of the base of a right circular cone, as input from the user.

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- Calculate the square of the radius, and store its value in a variable.
  (This value is used in all of the formulas. Use the stored value in all the formulas.)
    - Use the pre-defined **pow** function from the **cmath** library for squaring.
- Calculate the **slant height** of the right circular cone, using the above formula.
    - Use the pre-defined **sqrt** function from the **cmath** library to implement this calculation.
- Calculate the **surface area** of the right circular cone, using the above formula.
- Calculate the **volume** of the right circular cone, using the above formula.
- From the **main** function, display a few blank lines, and then a RESULTS header.
- From the **main** function, display the base length and height of the right circular cone, rounded to 2 decimal places as shown in the sample output. The decimals of both values should line up.
- Write and call a separate, **user-defined** function to output the calculation results:
    - The function should have thee input parameters: slant height, surface area, and volume
    - From within the user-defined function, display surface area, slant height, and volume of the right circular cone formatted as shown below, each rounded to 2 decimal places. The decimals of the three values should line up, and also line up with the input values displayed from main.
    - No calculations should be performed from within this function.

Sample Input and Output

```
Program for Right Circular Cone calculations

Enter cone radius (in inches): 1
Enter cone height (in inches): 2.2


RESULTS:
For a Right Circular Cone with
    Radius of             1.00 inches
    Height of             2.20 inches

Cone Slant Height is      2.42 inches
Cone Surface Area is     10.73 square inches
Cone Volume is            2.30 cubic inches
```

Documentation

1) Comment any constants/variables whose names are not ***completely*** descriptive.

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2) Be sure to include top of program comments as specified in course Content section 1.8:

Your name, the course, and the assignment number
What the Program Does including:
What is Input
Processing Performed
What is Output

3) You must also include a comment above **each user-defined function** that includes:
What the function does
The ***name*** and a ***description*** of any parameters that are input to the function

4) For readability, make sure you have:
- a SPACE before and after each operator (+, -, *, /, =)
- a SPACE after each comma in your code
- blank lines between ***sections*** of your code

5) Remember that dividing one integer by another integer will produce an integer result.
Example: 1 / 3 = 0
So if you want floating point results, be sure that one of the operands is a floating point value.
Example: 1 / 3.0 = 0.3333…

Submission

This programming assignment is due by midnight Sunday. Submit your program source code (both of the **.cpp** files) to the **Prog Assn 2** dropbox (located under the **Dropbox** tab in the online course). **Both** files should be attached to **one** submission for programming assignment 2.

Before submitting your program files, you MUST name them as follows:

```
Lastname-wk2-prog1.cpp
Lastname-wk2-prog2.cpp
```

For example:

```
Smith-wk2-prog1.cpp
Smith-wk2-prog2.cpp
```

Grading

The rubric that will be used to grade your program is included on the next pages.

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CS 361 Control Structures
Programming Assignment 2 Rubric

Program 1: (50 pts)				
Rating **Rating Category**	**Exemplary**	**Proficient**	**Basic (needs work)**	**Not Demonstrated**
Documentation	Top of program comments include: Student name, course, and assignment number, and clear program description with inputs, processing, and outputs listed. Top of function comments explain what function will do. Inline comments are included as necessary.	Missing one required component OR program description is too vague or missing some minor details about the input, output, or processing OR top of function comments too vague.	Documentation has multiple problems with missing, incomplete and/or incorrect items.	Only a few (or no) comments in program.
Data Storage Points	Constants defined for all fixed values. Variables all defined locally, using appropriate data types. Follows coding standards naming conventions and used descriptive names for all identifiers.	A few minor errors in data typing or assignment OR a few minor errors in identifier names OR missing necessary constant OR defined one global variable.	One or more major errors, OR did not follow any of the naming conventions for identifiers, OR defined multiple global variables.	Constants not used, identifier names are not descriptive, AND/OR there are multiple errors in the data types assigned.
Header Function Implementation	Header function correctly defined and called, and displays header correctly, as specified by the requirements.	Minor error in function definition or function call, OR output not displayed as specified.	Multiple errors in function definition and/or function call.	Header was not implemented as a function. It was simply output from within the main function.
Program Input	Displays descriptive prompts for each input indicating exactly what the user needs to provide, and reads the required input correctly.	Reads the required input correctly, but does not display an exacting prompt.	Prompts, but fails to read correctly some of the required input.	No prompt or data read from user. Hardcoded values used or program fails.
Program Logic	The program logic accomplishes all program requirements. Mathematical equations compute the correct results.	All mathematical equations compute the correct results, but the program logic does not accomplish all of the program requirements.	One or more of the mathematical equations compute incorrect results, AND the program logic does not accomplish all of the program requirements.	The computations and program logic accomplish few or none of the program requirements.

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Program Output	Displays all of the required output and follows required output format, according to program requirements	Displays all the required output, but there are one or two minor errors in the output format.	Does not display some of the required output, OR displays all of the required output, but has multiple errors in the output format.	Displays completely incorrect output, or does not display any output at all.
C++ Constructs / Readability / Miscellaneous	Demonstrates understanding of program structure. Code is organized and easy to follow, with no issues. There is no unnecessary code and code compiles correctly.	Code is fairly easy to read, but there are some spacing, indentation, and/or other minor issues, or extraneous code. Code compiles correctly.	Code is difficult to read or has other major issues AND/OR one small error prevents the code from compiling.	Multiple errors prevent the code from compiling AND/OR the code is unreadable.
Program 2: (50 pts)				
Documentation	Top of program comments include: Student name, course, and assignment number, and clear program description with inputs, processing and outputs listed. Top of function comments explain what function will do, and describe all input parameters. Inline comments are included as necessary.	Missing one required component OR program description is too vague or missing some minor details about the input, output, or processing OR top of function comments too vague.	Documentation has multiple problems with missing, incomplete and/or incorrect items.	Only a few (or no) comments in program.
Data Storage Points	Constants defined for all fixed values. Variables all defined locally, using appropriate data types. Follows coding standards naming conventions and used descriptive names for all identifiers.	A few minor errors in data typing or assignment OR a few minor errors in identifier names OR missing necessary constant OR defined one global variable.	One or more major errors, OR did not follow naming conventions for ANY of the identifiers, OR defined multiple global variables.	Constants not used, identifier names are not descriptive, AND/OR there are multiple errors in the data types assigned.
Program Input	Displays descriptive prompt for input indicating exactly what the user needs to provide, and reads the required input correctly.	Reads the required input correctly, but does not display an exacting prompt.	Prompts, but fails to read the required input, and uses hardcoded values.	Program fails because no input was read.

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Program Logic	The program logic accomplishes all program requirements. Mathematical equations compute the correct results.	All mathematical equations compute the correct results, but the program logic does not accomplish all of the program requirements.	One of the mathematical equations compute incorrect results, AND the program logic does not accomplish all of the program requirements.	Multiple mathematical equations compute incorrect results, and the program logic accomplishes few or none of the program requirements.
Output Function Implementation	Correct function definition (6) and function call (4), with proper parameter passing.	One or two minor errors in the function definition, function call, or with the parameter passing.	Multiple errors in the function definition, function call, and/or with the parameter passing.	Output was displayed from within the main function, instead of from a separate function.
Output	Displays output correctly, formatted as specified by the requirements.	There is a minor error in the output content or format.	Multiple errors in the output content or format.	Does not display any output.
C++ Constructs / Readability / Miscellaneous	Demonstrates understanding of program structure. Code is organized and easy to follow, with no issues. There is no unnecessary code and code compiles correctly.	Code is fairly easy to read, but there are some spacing, indentation, and/or other minor issues, or extraneous code. Code compiles correctly.	Code is difficult to read or has other major issues AND/OR one small error prevents the code from compiling.	Multiple errors prevents the code from compiling AND/OR the code is unreadable.

NOTE: Deductions for late submissions will be applied (-2% per day late)
But programs will **not be accepted** if they are submitted **more than 5 days** late (grade will be 0).

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